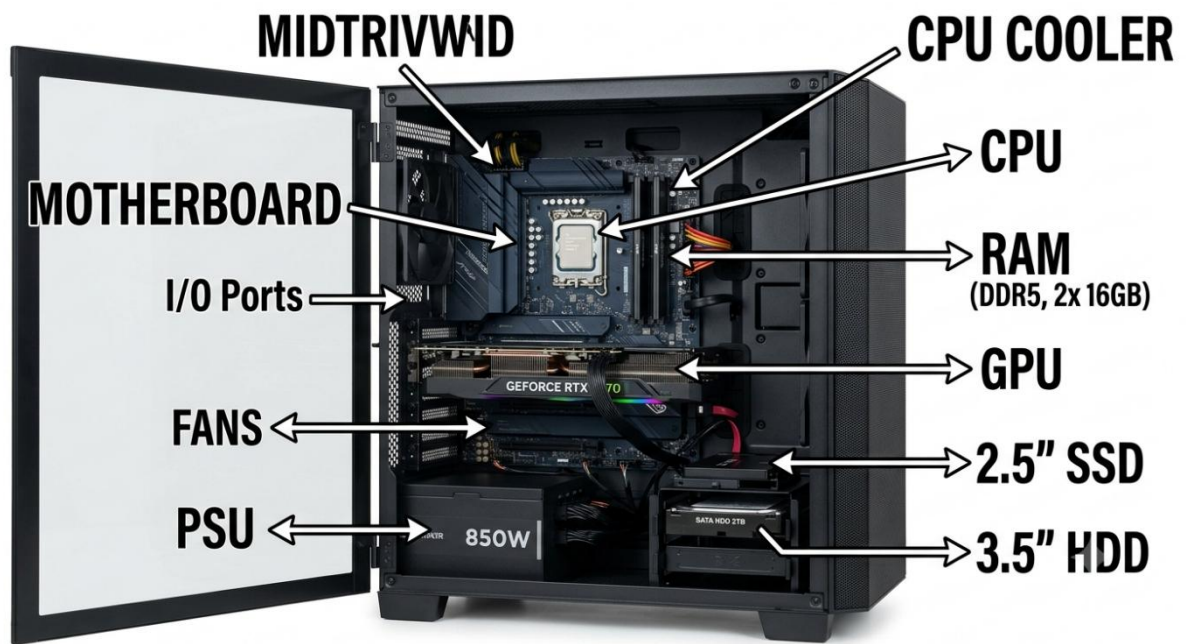


1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower, with moving mechanical parts.	Much faster, using flash memory.
Durability	More prone to physical damage.	Very durable and shock-resistant.
Use Cases	High-capacity storage, backups.	Gaming PCs, laptops, quick access files.

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Central Processing Unit)

The CPU acts as the brain of your PC, handling the game's core logic, physics calculations, and artificial intelligence. When playing battle royale games like PUBG or Free Fire, it constantly processes player positions, bullet trajectories, and map data. Without a fast processor, the game will struggle to track real-time actions, leading to stuttering and hitching.

RAM (Random Access Memory)

RAM serves as the super-fast, short-term storage that holds all the active game data currently in use. It allows your computer to instantly pull up

textures, player models, and map assets as you move across the battlefield. If your system runs out of RAM while playing, the game is forced to use your slower storage drive, causing massive lag spikes and slow loading times.

GPU (Graphics Processing Unit)

The GPU is entirely dedicated to rendering the visuals, 3D geometry, and special effects you see on your screen. It takes the heavy graphical workload off the CPU to draw the complex environments, lighting, shadows, and character skins at a high frame rate. A powerful graphics card ensures that fast-paced action remains smooth and visually crisp, giving you a competitive edge.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

To run the latest version of the franchise (such as *EA Sports FC 26*, the continuation of the classic FIFA series) smoothly at high settings and frame rates, you should focus your upgrade strategy on four core components:

1. Graphics Card (GPU)

- **Why Upgrade:** The latest football simulations utilize demanding graphical physics engine upgrades—handling advanced features like strand-based realistic hair simulation, cloth geometry for jerseys, and dense 3D crowds. While the recommended spec starts around a GTX 1660 or RX 5600 XT for baseline 1080p gaming, upgrading to a modern mid-range card (like an RTX 4060 or RX 7600) will ensure fluid 60+ FPS gameplay at High/Ultra settings, or even allow you to scale up to 1440p/4K resolutions smoothly.

2. Processor (CPU)

- **Why Upgrade:** Football titles are surprisingly CPU-intensive because the computer has to process complex tactical positioning, smarter referee/goalkeeper AI behaviors, and real-time collision ball physics for 22 active players simultaneously. Moving beyond older 4-core CPUs to a modern 6-core or 8-core processor (such as an AMD Ryzen 5 7600X or Intel Core i5-14600K) prevents localized menu freezing and eliminates micro-stuttering during fast-paced counterattacks.

3. RAM (Memory)

- **Why Upgrade:** The game requires a recommended baseline of 12 GB of RAM just to function comfortably alongside Windows. Upgrading your system to **16 GB or 32 GB of dual-channel RAM** gives your PC the breathing room to instantly pull stadium assets and player models. This headroom prevents the sudden, frustrating frame drops that can happen right as you enter the penalty box or load into an intense Ultimate Team match.

4. Storage (SSD)

- **Why Upgrade:** Modern entries occupy around **100 GB of storage space**. Running this game on an old mechanical HDD will subject you to agonizingly long loading screens between menu transitions and match cutscenes. Upgrading to a fast **NVMe M.2 SSD** ensures the game boots up in seconds, menus glide instantly, and matchmaking transitions are entirely seamless.